

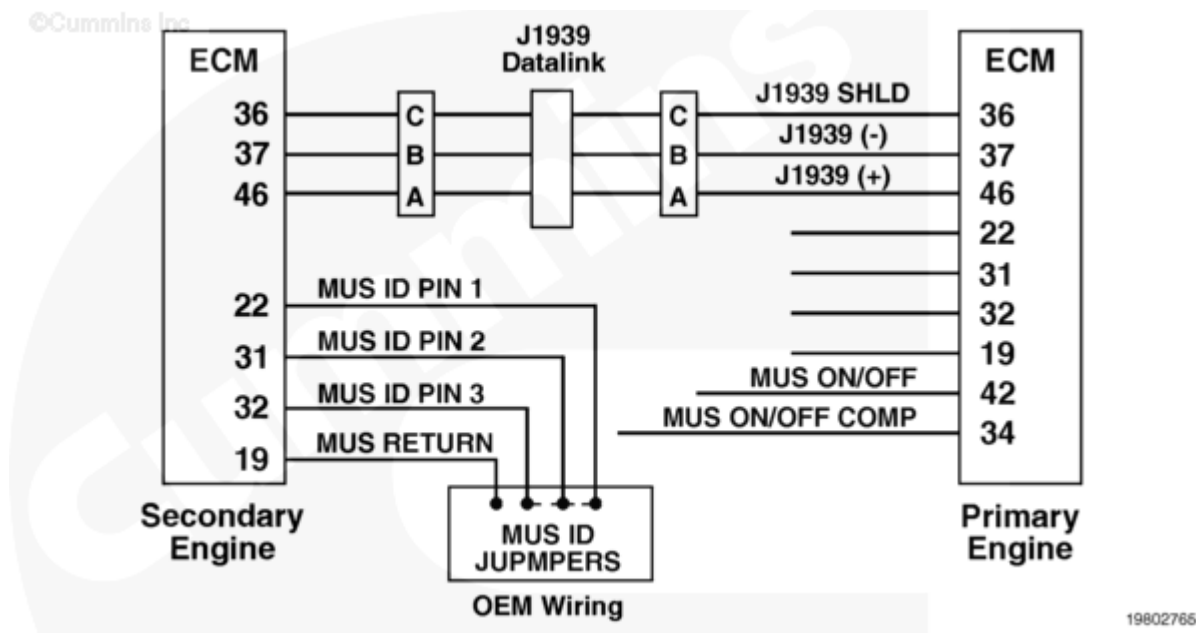
Fault Code: 237

Multiple Unit Synchronization (soft-coupled Marine)

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 237 PID(P): S030 SPN: 644 FMI: 2/2 Lamp: Yellow SRT: 00-395	Duty cycle of the throttle input signal to the primary or secondary engine for multiple unit synchronization is less than 3 percent or more than 97 percent.	All engines (primary and secondary) are shut down with increasing time after alert if hard-coupled. Only secondary engines are shut down with increasing time after alert if soft-coupled.



Multiple Unit Synchronization Circuit, Soft-Coupled Marine

Circuit Description

The primary engine broadcasts the engine speed control message over the SAE J1939 datalink to all secondary engines. Jumper wires are used to configure an engine as a primary engine or as a secondary engine. **Only** one engine can be configured as the primary engine. Up to 5 engines can be configured as secondary engines on the same SAE J1939 datalink.

Component Location

The SAE J1939 datalink and jumper wire connections are part of the OEM harness. Location of the jumper wire connections vary by the OEM. Refer to the OEM manual for component location.

Shoptalk

The configuration of the 3-pin inputs to the electronic control module (ECM) on the OEM harness determines if an engine is a primary or secondary engine. See the matrix provided. To activate a pin, the pin **must** be grounded. To deactivate a pin, the pin **must** be an open circuit.

Pin 3	Pin 2	Pin 1	Primary/Secondary
INACTIVE	INACTIVE	INACTIVE	Default to Primary
INACTIVE	INACTIVE	ACTIVE	Primary
INACTIVE	ACTIVE	INACTIVE	Secondary 0
INACTIVE	ACTIVE	ACTIVE	Secondary 1
ACTIVE	INACTIVE	INACTIVE	Secondary 2
ACTIVE	INACTIVE	ACTIVE	Secondary 3
ACTIVE	ACTIVE	INACTIVE	Secondary 4
ACTIVE	ACTIVE	ACTIVE	Default to Primary

Verify that the keyswitch is on for all multiple unit synchronization engines. Verify that no throttle faults are active.

Verify that the SAE J1939 datalink is active and communicates correctly with INSITE™ electronic service tool on all engines. If any ECM will not communicate over the SAE J1939 datalink. Refer to Procedure 022-999 in Section F. (</qs3/pubsys2/xml/en/procedures/00/00-022-999.html>)

Refer to Troubleshooting Fault Code t05-237 (</qs3/pubsys2/xml/en/procedures/82/82-t05-237.html>)